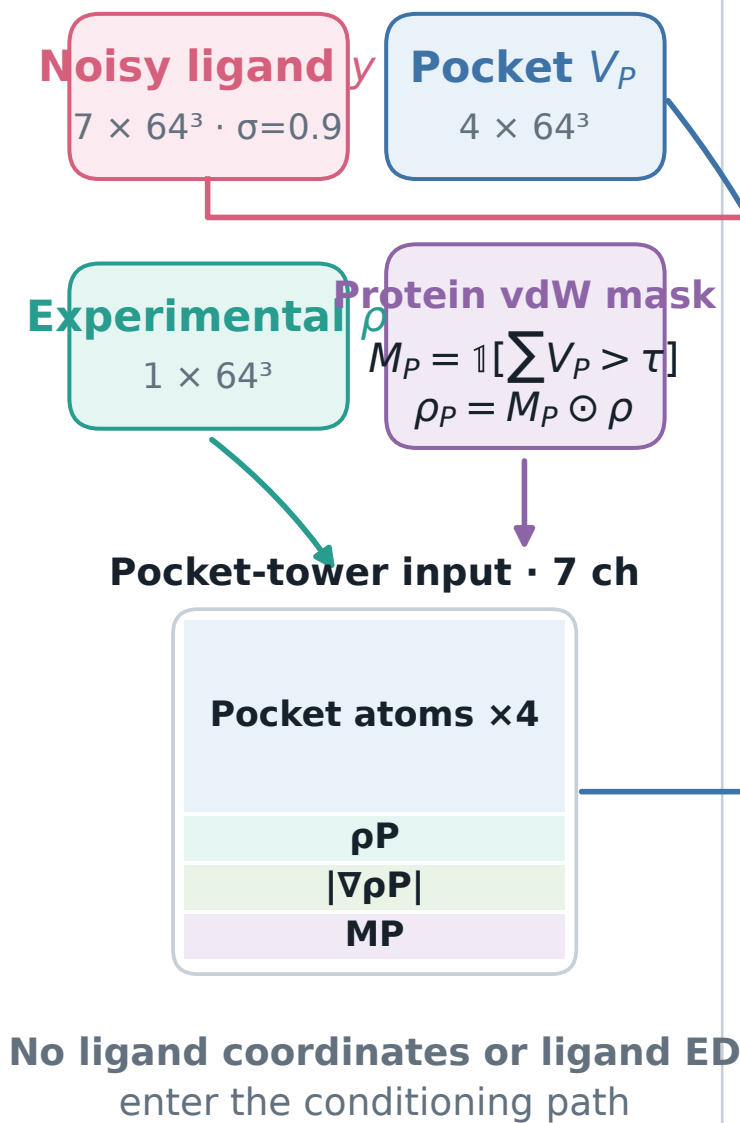


Transfer of the pretrained pocket encoder into VoxBind

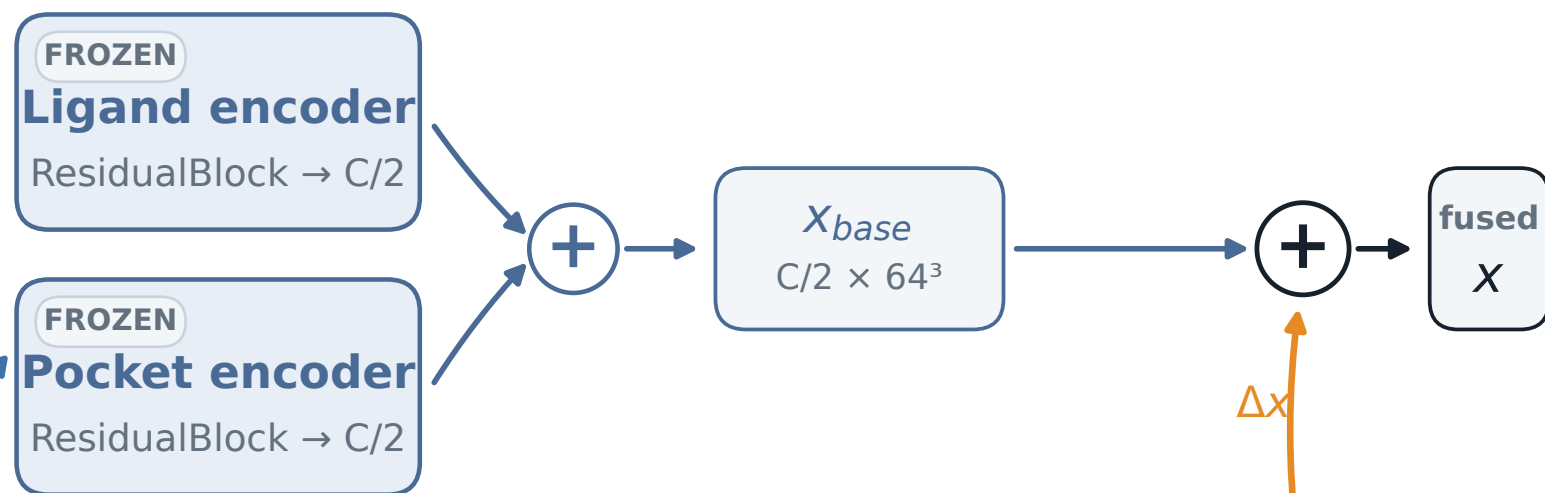
A zero-initialized spatial adapter adds experimental-ED context while the pocket tower and VoxBind remain frozen

(a) Conditions

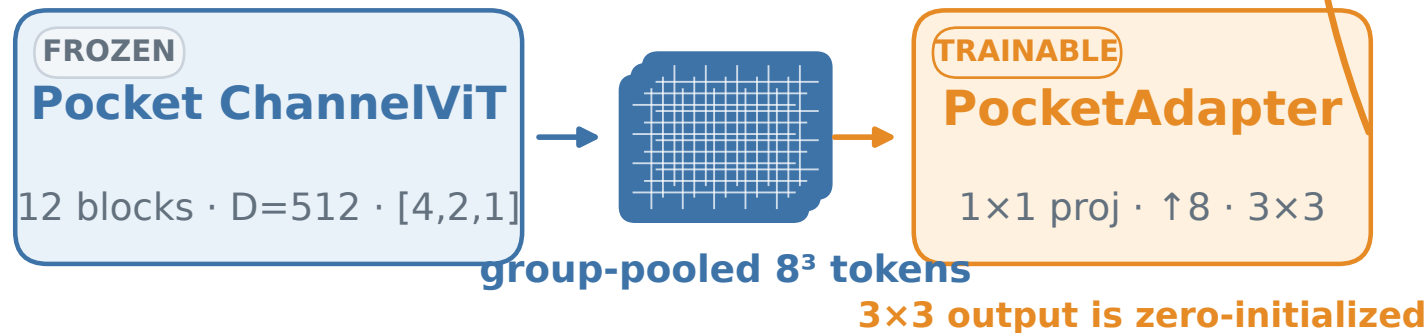


(b) Frozen backbone + trainable residual

Original coordinate path



Transferred experimental-ED path



UPDATED PocketAdapter only

Frozen: both coordinate encoders · pocket ChannelViT · 3D U-Net · output head

(c) Denoising and sampling



Adapter training
 $\|\hat{x} - x_{clean}\|^2$
gradients $\rightarrow$ adapter only

Generation
 $s_\theta = (\hat{x} - y)/\sigma^2$
conditional score

Walk-jump sampling
Langevin walk $\rightarrow$ denoising jump

At initialization, $\Delta x=0$: the integrated model exactly reproduces the frozen VoxBind baseline.